MAE 3127
Fluid Mechanics Laboratory

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Topics covered today:

1. Gates
2. Bernoulli's equation
3. Conservation of momentum

School of Engineering & Applied Science
THE GEORGE WASHINGTON UNIVERSITY

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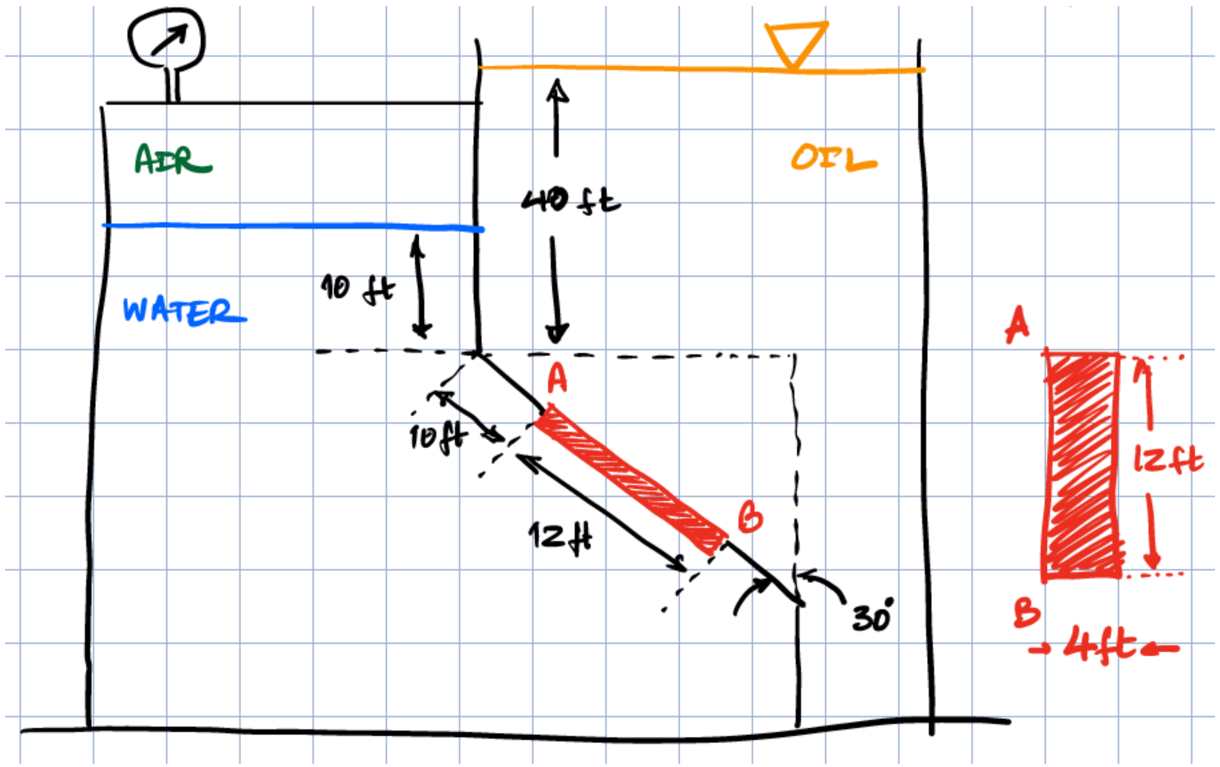
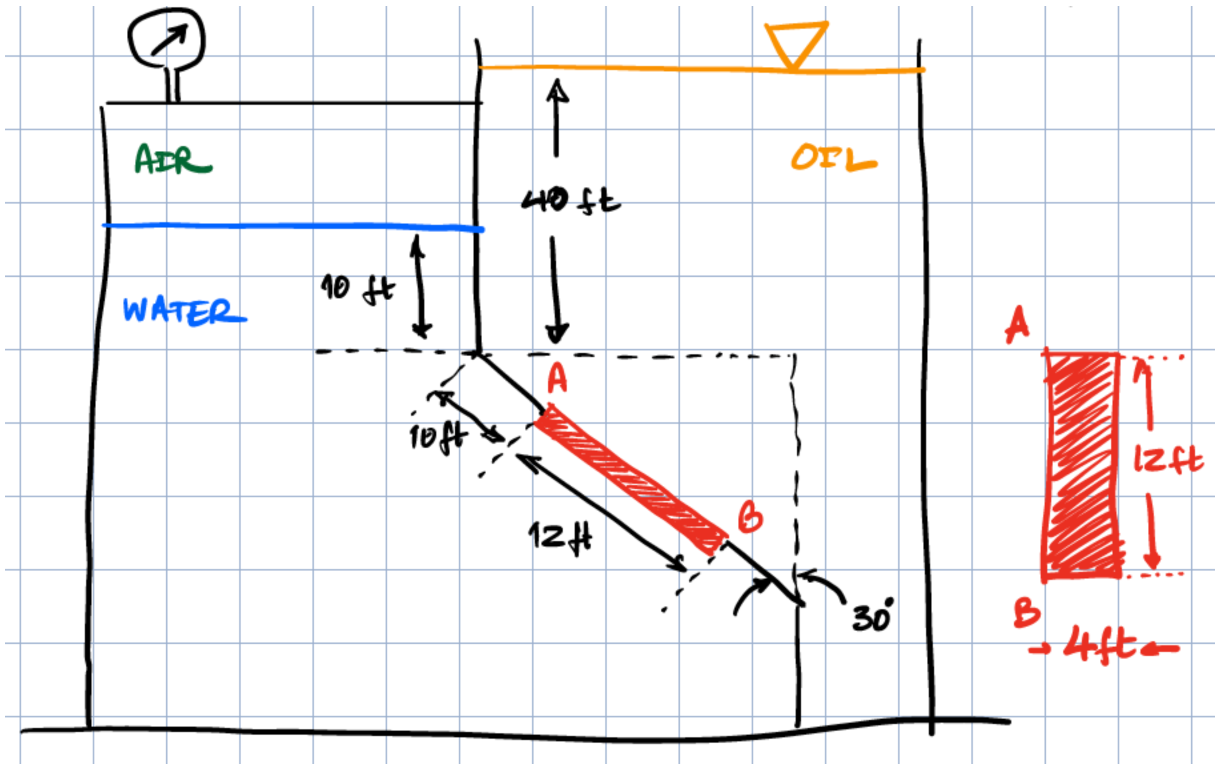
Photo: Kartik Bulusu



KEY IDEAS

↳ Two fluids on either side of gate AB
LEFT & RIGHT

CALCULATION	LEFT SIDE	RIGHT SIDE
Location: Center of gravity	h_{cL}	h_{cR}
Resultant force	F_{RL}	F_{RR}
Location: center of pressure	$(y_R - y_c)_L$	$(y_R - y_c)_R$



↳ Overall Resultant force (F_R)

$\boxed{F_R = \text{Difference of } F_{RL} \text{ \& } F_{RR}}$

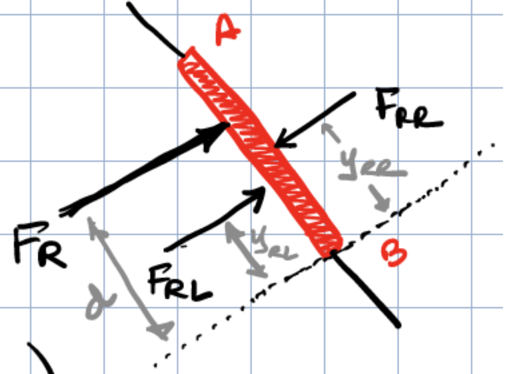
↳ F_R, F_{RL} & F_{RR} create a moment on gate AB

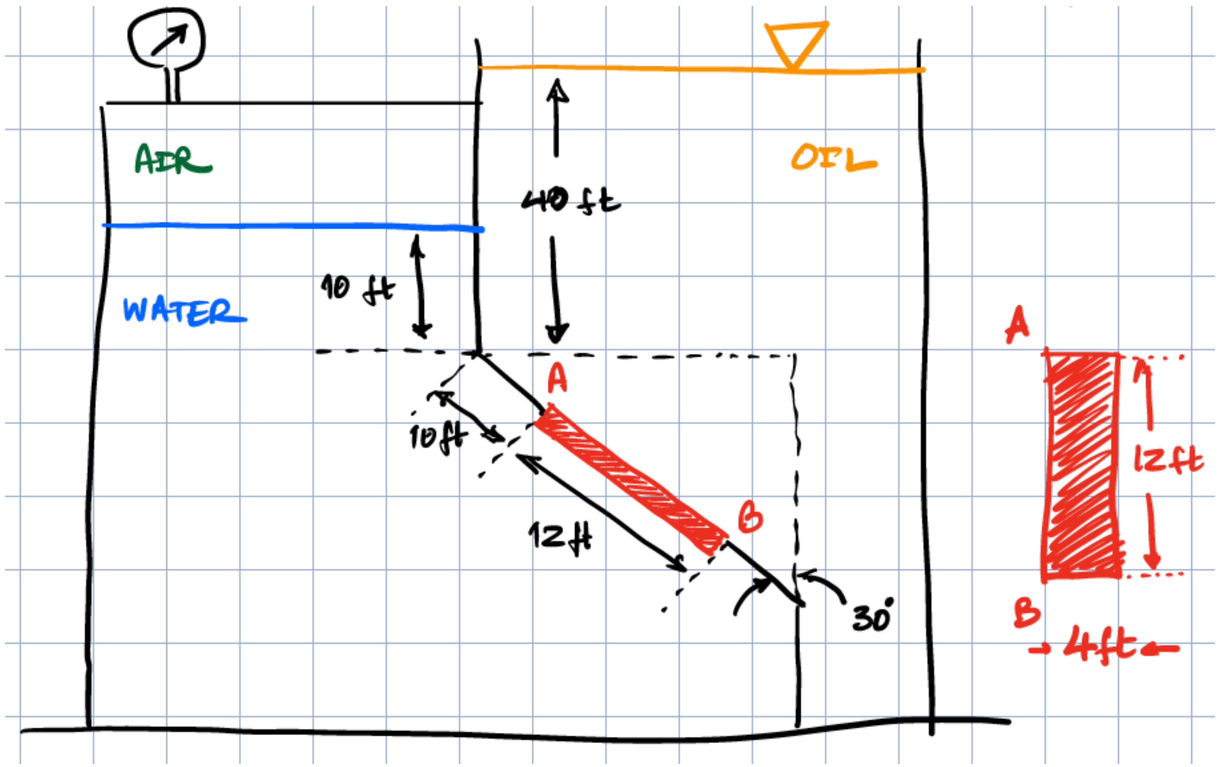
* All distances are from B

y_{RR} → From (B) to (center of pressure on Right side)

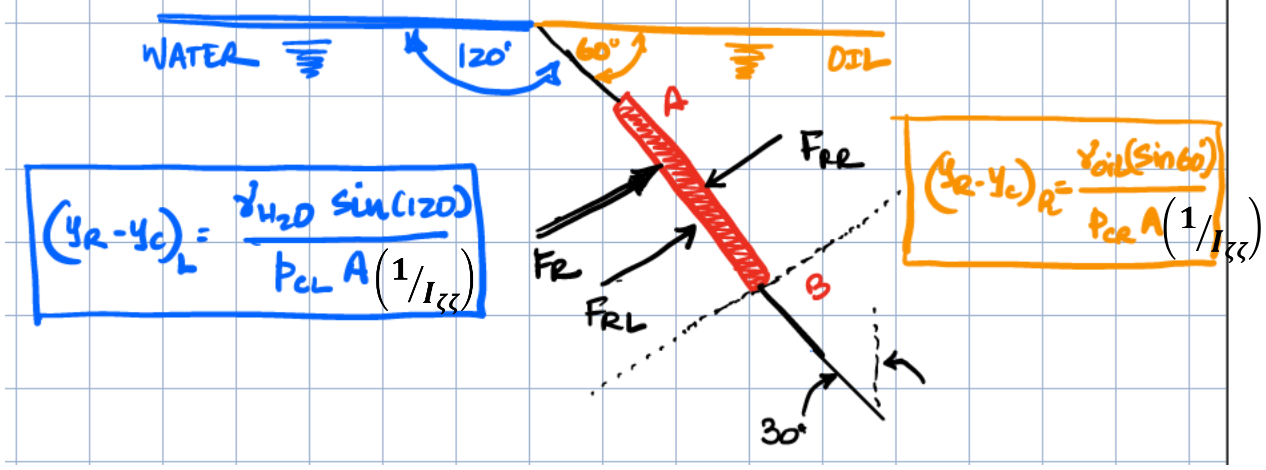
y_{RL} → from (B) to (center of pressure on Left side)

d → From (B) to (Overall Resultant force Location)





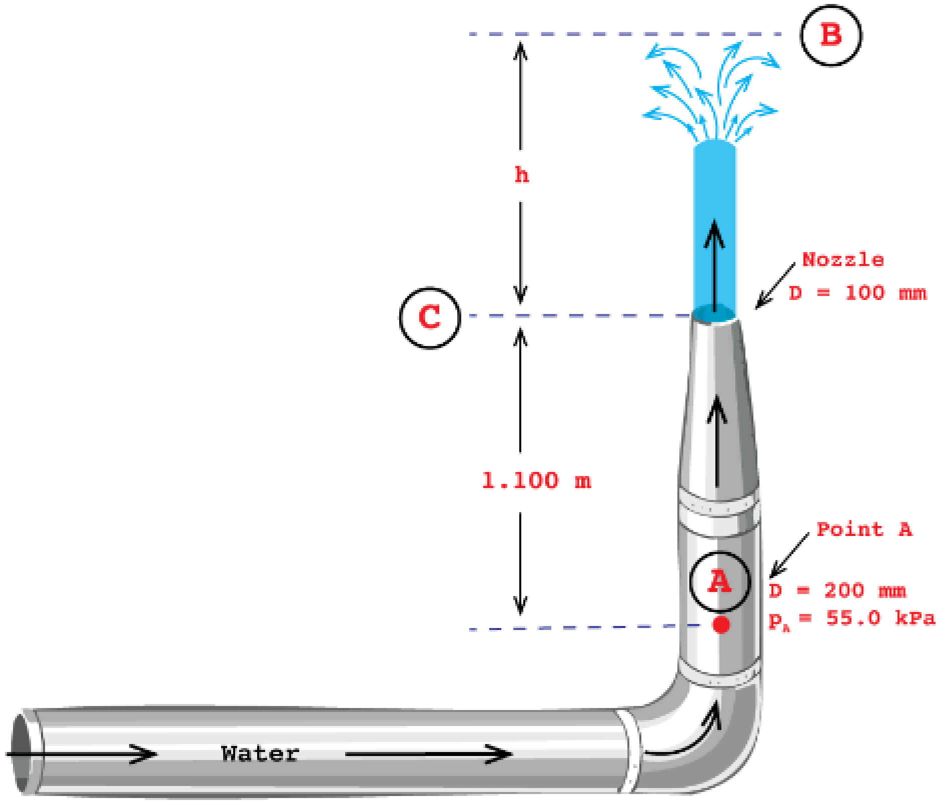
↳ Note on center of pressure location



$$\boxed{(y_R - y_c)_L = \frac{\gamma_{H_2O}\sin(120)}{p_{cL}\, A\left(1/I_{\zeta\zeta}\right)}}$$

$$\boxed{(y_R - y_c)_R = \frac{\gamma_{oil}(\sin 60)}{p_{cR}\, A\left(1/I_{\zeta\zeta}\right)}}$$

R2 P2: Determine the height of water H that will shoot out the nozzle shown below. Pressure at point A is 55.0 kPa (gage). Assume that the fluid is frictionless and that the jet does not spread after is leaves the nozzle. (*Hint: note that the water at the top of the jet has zero velocity when it reaches height h.*)



Background Information:

For a fluid particle moving along a streamline, **Bernouilli's equation (based on conservation of energy)** states that:

$$p + \frac{1}{2}\rho v^2 + \rho g h = constant$$

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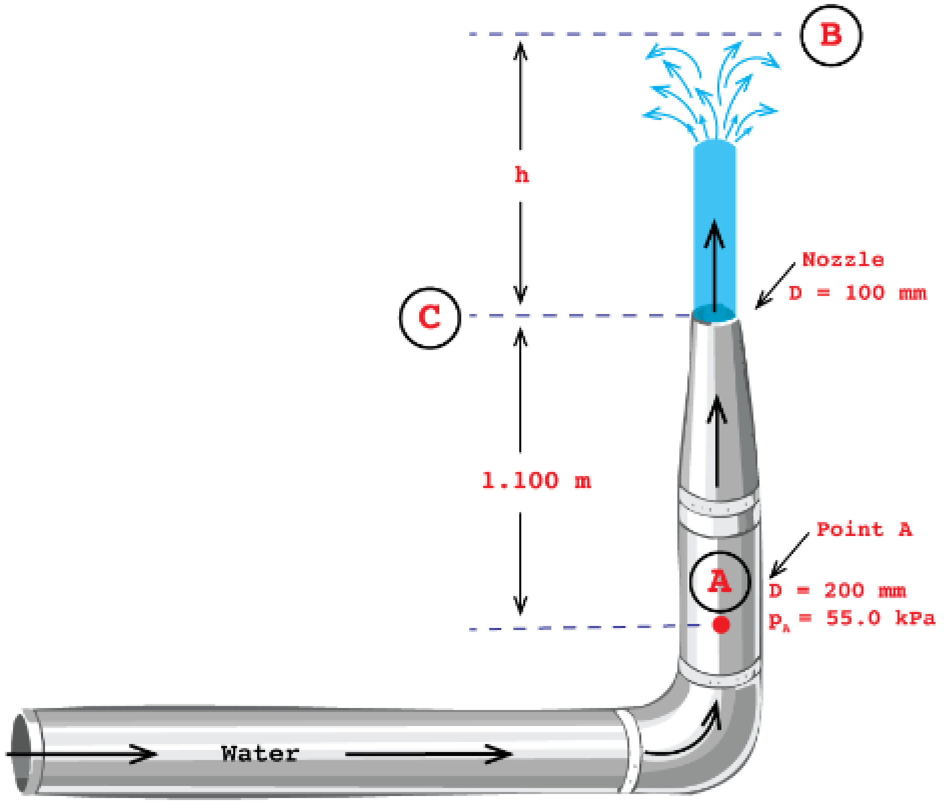

Background Information:

Bernouilli Equation Assumptions:

- steady
- incompressible
- frictionless flow

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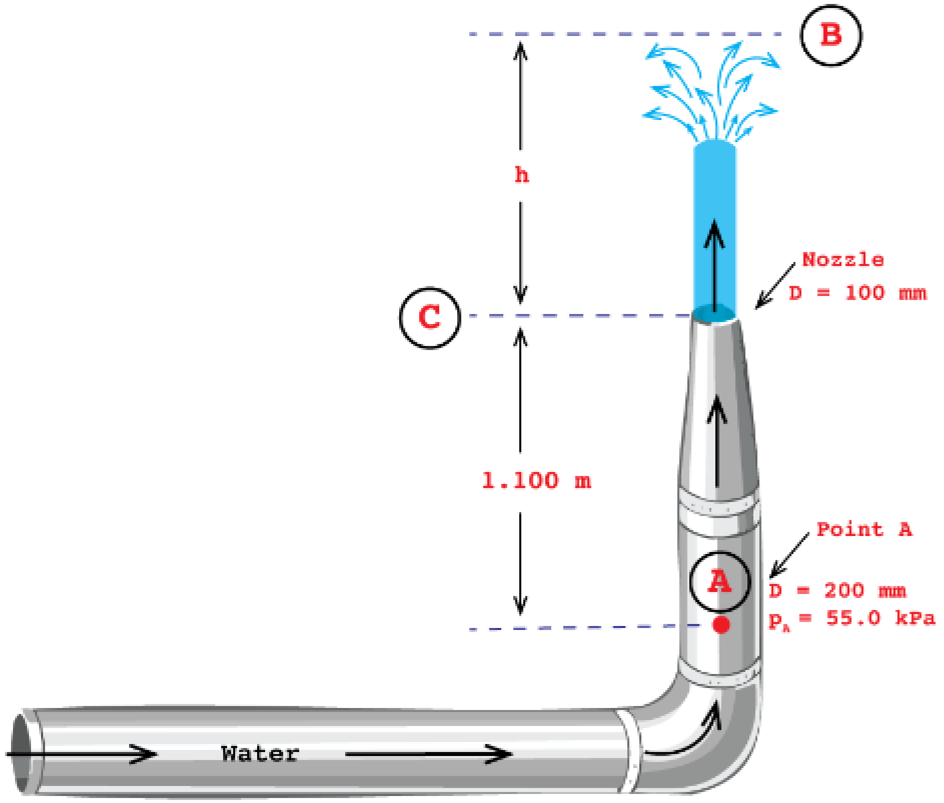

Background Information:

Conservation of mass:

$$\dot{m} = \rho v A = constant$$

$\dot{m}$: mass flow rate [kg/s]
ρ : density of the fluid [kg/m^3]
v : velocity of the fluid [m/s]
A : cross-sectional area of the flow [m^2]

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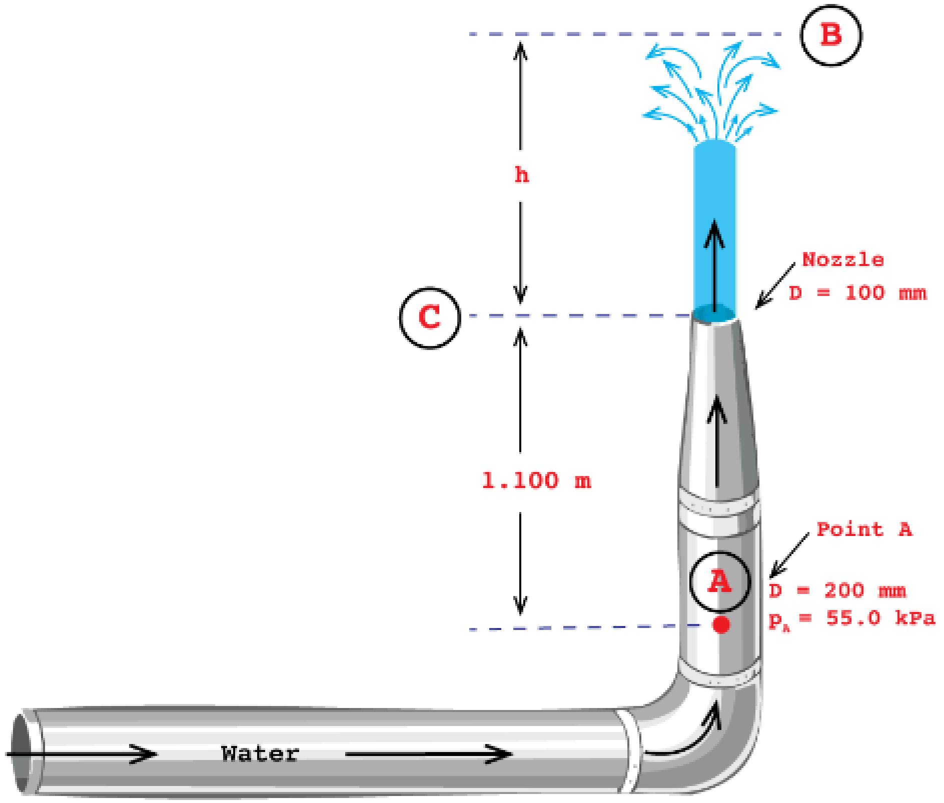

Strategy:

- Energy is conserved along the streamline
- We can solve the Bernouilli Equation for Points A, B, C:

1. Apply Bernouilli's equation from A to B
2. Apply Bernouilli's equation from A to C
3. Use conservation of mass